211

NUCLEAR DATA SHEETS A = 35 Vol. 211 (2026) 1–422

ELSEVIER

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Volume 211, June 2026

ISSN 0090-3752

Nuclear Data Sheets

Produced by

The National Nuclear Data Center
Brookhaven National Laboratory
Upton, NY 11973-5000, USA

for

The International Network of
Nuclear Structure and Decay Data Evaluators

NOTE

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NUCLEAR DATA SHEETS

A Journal Devoted to Compilations and Evaluations of Experimental and Theoretical Results in Nuclear Physics

Contents of Recent Issues

A-Chains in Press or Accepted for Publication
A = 57, 130, 72, 74

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0090-3752(202606)211:C;1-M

Nuclear Data Sheets Symbols and Abbreviations

A	mass number*, A = Z + N
A_2, A_4	coefficients of Legendre polynomials in angular-correlation or angular-distribution measurement
av	average
B(EL), B(ML)	reduced EL, ML transition probability in $e^2 \times (\text{barn})^L$, $\mu_N^2 \times (\text{barn})^{L-1}$
calc, CA	calculated, calculation
CCBA	coupled-channel Born approximation
ce	conversion electron
chem	chemical separation
circ	circular
c.m.	center of mass
coef	coefficient
coin	coincidence
Coul. ex.	Coulomb excitation
CP	circular polarization
cryst	crystal-diffraction spectrometer
C^2S, C^2S'	one-nucleon spectroscopic strength for pickup, stripping reactions
d	day
D	dipole
DSA	Doppler shift attenuation
DWBA	distorted-wave Born approximation
DWIA	distorted-wave impulse approximation
E	energy
E(ε)	energy of electron-capture transition (endpoint of γ-continuum + K-electron separation energy of daughter)
E1, E2, EL	electric dipole, quadrupole, 2^L-pole
excit	excitation function
expt	experiment, experimental
F	fission
F-K	Fermi–Kurie (plot)
FWHM	energy resolution, full width at half maximum
g	gyromagnetic ratio*
GDR	giant dipole resonance
GQR	giant quadrupole resonance
g.s.	ground state
h	hour
H	magnetic field
HF	hindrance factor
hfs	hyperfine structure
HI	heavy ion
I	intensity
IAR	isobaric analog resonance
IAS	isobaric analog state
IBS	internal bremsstrahlung spectrum
IMPAC	ion implantation perturbed angular correlation technique
inel	inelastic
ion chem	chemical separation by ion exchange
IT	isomeric transition
J	total angular momentum quantum number*
K	projection of nuclear angular momentum J on nuclear symmetry axis
K, L, M	K-, L-, M-shell internal conversion
K/L	K-, L-conversion electron ratio
L	(1) orbital angular momentum quantum number*, (2) multipolarity
L(n), L(p)	L-transfer in neutron, proton transfer reaction
min	minute
M+	$M + N + O + \cdots$
M1, M2, ML	magnetic dipole, quadrupole, 2^L-pole
mag spect	magnetic spectrometer
max	maximum
Moss	Mossbauer effect
ms	(1) mass spectrometer, (2) millisecond
mult	multipolarity/character
N	neutron number*, N = A − Z
NMR, NQR	nuclear magnetic, quadrupole resonance
norm	normalization
PAC	perturbed angular correlation
pc	proportional counter
p, $\gamma(\theta)$	angular distribution of γ-rays with respect to a proton beam
p, γ(t)	time distribution of photons with respect to a pulsed proton beam
pol	polarized, polarization
priv comm	private communication
PWBA	plane-wave Born approximation
Q	(1) reaction energy*, (2) disintegration energy*, (3) quadrupole moment*, in units of barns, (4) quadrupole
Q(ε)	total disintegration energy in ε decay
Q($\beta-$)	total disintegration energy in $\beta-$ decay
Q(α)	total disintegration energy in α decay, E(α) + E(recoil)
R	$r_o A^{1/3}$, nuclear radius*
RDM	recoil distance measurement
RUL	recommended upper limit for γ-ray strength
rel	relative
res	resonance
s	second
S	spectroscopic factor
S′	$[(2J_f+1)/(2J_i+1)]$S
S(n) or S_n, S(p) or S_p	energy necessary to separate a neutron, proton from nucleus
scatt	scattering
scin	scintillation counter
semi	semiconductor detector
SF	spontaneous fission
spall	spallation
sr	steradian
syst, SY	systematics
t	triton
T	(1) isobaric spin, (2) temperature
Tz	Z-component of isobaric spin, $(N-Z)/2$
$T_{1/2}$	half-life*
th	thermal
thresh	threshold
tof	time-of-flight measurement
vib	vibrational
W.u.	Weisskopf single-particle transition speed
y	year
Z	atomic number*, Z = A − N
α	total γ-ray internal conversion coefficient $N(ce)/N(\gamma)$*
α(K), α(L)	γ ray internal conversion coefficient for electrons ejected from the K-, L-shell
$\alpha\gamma$, $\beta\gamma$, $\gamma\gamma$,	coincidences of α's and γ's, β's and γ's, γ's and γ's
$\alpha\gamma(\theta, H, t)$, $\beta\gamma(\theta, H, t)$, $\gamma\gamma(\theta, H, t)$	$\alpha\gamma$-, $\beta\gamma$-, $\gamma\gamma$-coincidences as function of angle, magnetic field, time
β_2, β_3, βL	quadrupole, octupole, 2^L-pole nuclear deformation parameter
$\beta\gamma$(pol), $\gamma\gamma$(pol)	polarization correlation of γ's in coincidence with β's, γ's
Γ, $\Gamma(\gamma)$, Γ(n)	level width*, partial width for γ-, n-emission
$\gamma(\theta, H, T)$	γ-intensity as function of angle, magnetic field, temperature
$\gamma\pm$	annihilation radiation
δ	ratio of reduced matrix elements of (L + 1)- to L-pole radiation with sign convention of Krane and Steffen, Phys. Rev. C2, 724 (1970)
ε	electron capture
εK, εL, εM	electron capture from K-, L-, M-shell
$\varepsilon(\gamma)$B(E2), ε(ce)B(E2)	partial B(E2) for photon, conversion electron detection
θ	indicates angular dependence
λ	(1) projection of particle angular momentum on nuclear symmetry axis, (2) radiation type, e.g., M1, M2, . . .
μ	magnetic moment of particle*, given in nuclear magnetons (μ_n)
ν	neutron shell-model configuration
π	parity, proton shell-model configuration
σ	cross section*
$\Sigma(\gamma\gamma)$	coincidence summing of γ-rays
ω(K), ω(L)	K, average-L fluorescence yield
%α	percent α branching from level
%$\beta-$	percent $\beta-$ branching from level
%$\beta+$	percent $\beta+$ branching from level
%ε	percent ε branching from level
%IT	percent (γ + ce) branching from level
%SF	percent spontaneous fission from level
$\langle r^2 \rangle$	root-mean-square of nuclear radius

Prefixes*

T	tera	$(=10^{12})$	μ	micro	$(=10^{-6})$
G	giga	$(=10^{9})$	n	nano	$(=10^{-9})$
M	mega	$(=10^{6})$	p	pico	$(=10^{-12})$
k	kilo	$(=10^{3})$	f	femto	$(=10^{-15})$
c	centi	$(=10^{-2})$	a	atto	$(=10^{-18})$
m	milli	$(=10^{-3})$			

Symbols for particles and quanta*

n	neutron	π	pion
p	proton	μ	muon
d	deuteron	e	electron
t	triton	ν	neutrino
α	α-particle	γ	photon

* Recommended by Commission on Symbols, Units, and Nomenclature of International Union of Pure and Applied Physics